

Higher National Diploma in Network Engineering - Year 2

CLOUD COMPUTING

Account guardrails lab



Day 1 - Afternoon

Running scenario: Thara Hospitals patient portal pilot

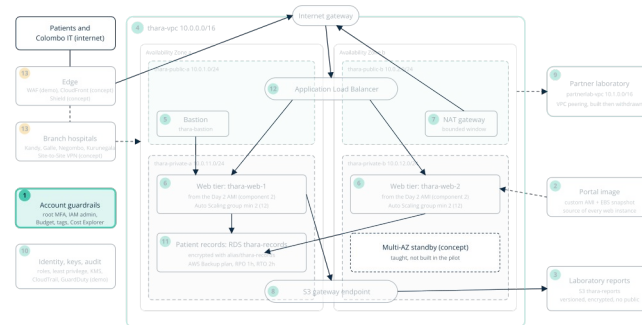
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Today

Before anything is built, the account is locked, watched and pinned to Mumbai.

- Root, admin, and why they differ - 15 min
- Budgets, credits and the ceiling - 15 min
- Shared responsibility, applied - 15 min



TODAY

1 Guardrails

CONCEPT 1

Root, admin, and why they differ

The Head of IT's personal login must not be the account's root.

15 min



IT

HEAD OF IT

If I leave tomorrow, or lose my phone, the hospital must still own its account. My personal login cannot be the key to everything.

The Head of IT, on why root is locked away and everyone works as a named user.

Root owns the account; the admin does the work

- The root user owns the account, and no policy can restrict it
- A named IAM admin user does daily work; root keeps a handful of account-level tasks
- MFA on **both** today; access keys on **neither**

YOU ALREADY KNOW THIS

The enable secret on a core switch versus a named operator account in your AAA server.

COMMON MISTAKE

MFA on root is enough.

The admin user needs it too, because it is the one you use every day.

CONCEPT 2

Budgets, credits and the ceiling

The CFO's actual-versus-forecast report starts with this budget.

15 min



CFO

CHIEF FINANCIAL OFFICER

Every month I want the actual against the forecast. A budget that reads zero because credits hide the spend tells me nothing.

The Chief Financial Officer, on what the budget is for.

A budget that includes credits never alerts

- Free Plan credits pay the bill, so a credit-inclusive budget reads zero
- Exclude **Credit** so the budget measures gross usage
- Alerts at USD 10, USD 25 and USD 50 are early warning, not enforcement
- Basic support is free and unlocks some Trusted Advisor checks

YOU ALREADY KNOW THIS

A bandwidth alert on a leased line emails you at 80 percent; it does not shape the traffic.

A budget alerts. It does not stop anything.



Open the budget-credit-trap visual and flip the credits switch before step 7.



COMMON MISTAKE

The budget stops spending.

It sends an email. Nothing in AWS stops spending except you.

CONCEPT 3



Shared responsibility, applied

The DPO will ask where the data is.

15 min



DATA PROTECTION OFFICER

Where will patient data be stored? I need a place I can name, not "the cloud".

The Data Protection Officer, on why the region is a responsibility.

Your first share of the Shared Responsibility Model



- IAM is the first customer responsibility; nothing AWS does protects a weak root password
- Region choice decides where data lives: every lab is pinned to **ap-south-1** (Mumbai)

The answer to the DPO is a region: Asia Pacific (Mumbai).

LAB - DAY 1

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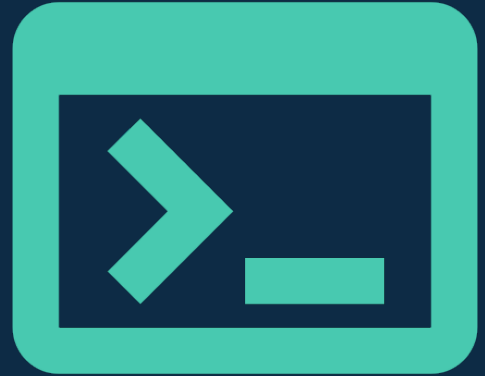
Account guardrails lab

You build:

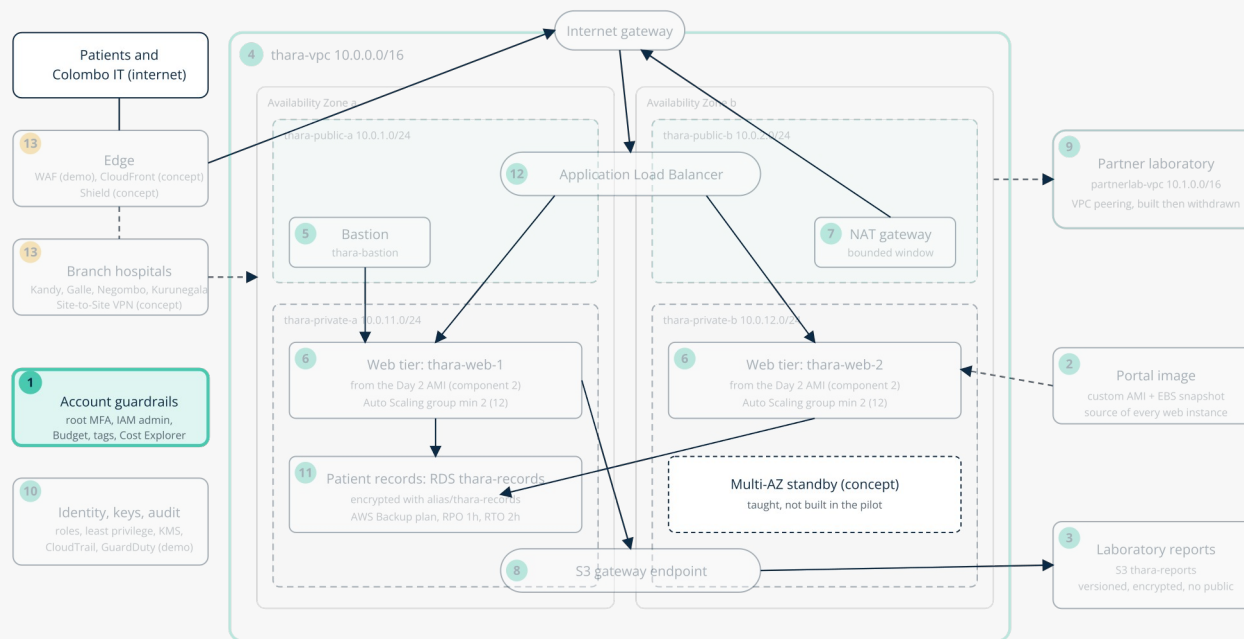
1

Estimated credit use: about USD 0.00

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The Thara architecture



SEGMENT 1 - 20 MIN



Activation triage

Cost: nil

Confirm the account is active



1 Sign in as root. Console Home should show the Explore AWS and Cost and Usage widgets.

Expected: A credit balance and the days left on your plan are shown.

Account still activating? Use the fallback IAM user you are issued today, and repeat the steps in your own account later.

SEGMENT 2 - 35 MIN



Root MFA and IAM admin user

Cost: nil

Secure root, then let IAM see billing



2

Security credentials, then Assign MFA device, then Authenticator app. If a device is already listed, verify it is yours.

Expected: One MFA device listed; the next root sign-in asks for a code.

3

Sign in as root again. Account, then IAM user and role access to Billing information, then Edit, Activate IAM Access, Update.

Expected: The page confirms IAM access to billing is activated.

Skip step 3 and your admin user gets access denied in Budgets and Cost Explorer.

Create the admin user and leave root behind



4 IAM, Users, Create user `admin-<student id>`: console access, I want to create an IAM user, Attach policies directly, AdministratorAccess.

Expected: The user is listed. Copy the IAM sign-in URL from the dashboard.

5 Assign MFA to the admin user, then sign out of root for the rest of the module.

Expected: The admin user shows one MFA device.

6 Sign in with the IAM URL as the admin user and set the region to ap-south-1.

Expected: An MFA prompt, then Asia Pacific (Mumbai) in the top bar.

SEGMENT 3 - 35 MIN



Budget and Cost Explorer

Cost: nil

Create a budget that measures gross usage



7

From Explore AWS, open the cost budget activity: Customize (advanced), Cost budget, `thara-budget`, Monthly, Fixed, 50.00. Under Budget scope add a Charge type filter that excludes Credit. Alerts on Actual spend at 10, 25 and 50, absolute value, to your email.

Expected: `thara-budget` listed: USD 50.00 budgeted, USD 0.00 actual.

Use Actual, not Forecasted. A new account has no forecast for weeks.

Switch on Cost Explorer



8

Cost Explorer, then Launch Cost Explorer. Then open the Free Tier page and the Credits page.

Expected: Data is being prepared, up to about a day. Your credit is listed.

SEGMENT 4 - 25 MIN



Trusted Advisor and ledger

Cost: nil

Trusted Advisor and the first ledger row



9 Open Trusted Advisor, open Security, choose Refresh, and note which checks are free on Basic.

Expected: MFA on root account shows no problem.

10 Ledger row for `thara-budget`: day 1, Guardrails, ap-south-1, account id with the middle digits masked.

Expected: One complete row, estimated cost 0.00.

SEGMENT 5 - 20 MIN



Evidence capture

Cost: nil

Prove it, then break it



PROVE IT

- A private window sign-in as the admin user asks for MFA
- `thara-budget` shows a Charge type filter excluding Credit
- Cost Explorer reopens without its welcome page
- Trusted Advisor's MFA on root account check is clear

BREAK IT

- Remove the Credit exclusion; the filter disappears and actual still reads USD 0.00
- Predict the reading at USD 12 of gross usage, check it in the visual, restore

Tear down and record



TEARDOWN

- Nothing to delete. In ap-south-1: no instances, only the default VPC, no buckets
- Keep the admin user, both MFA devices, the budget and Cost Explorer

COST

Estimated credit use: about USD 0.00

- Write the day's total against the estimate in your ledger

FOR YOUR EVIDENCE PACK

- Screenshot of the IAM user with MFA enabled (mask the account id's middle digits)
- Screenshot of the Budget filter showing Credit excluded and the three alert thresholds
- First cost ledger row

Before the next session



READ

IAM User Guide

"Security best practices": root, MFA and access keys.

SELF-CHECK

CLF-C02 bank

Domain 4 set A, 10 recall items.

TASK

Evidence Pack 1

Due before the next teaching day. Keep working on PQ1.

Questions & Discussion

Next: the portal's first server, its image, a container beside it, and the reports bucket.



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